Practical 14

AIM: write a code using RAW sockets to implement packet sniffing

Algorithm:

1. open a raw socket in promiscous mode
2. Receive a packet (raw bytes) from the socket
3. parse Ethernet header to get Ethertype.
4. If Ethertype == IPv4, parse IP header to get protocol src IP, dstIP
5. optionally parse TCP/UDP/ICMP headers based on IP protocol
6. Display log the packet info repeat to keep sniffing.

Input:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet)
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        # determine the protocol
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
```

```
elif
    protocol_name = UDP

else
    protocol_name = "unknown protocol"
# print packet details
    print(f" protocol :{protocol_name}")
    print(f" Source IP : {src_ip}")
    print(f" Destination IP :{dst_ip}")
    print("--"*50)

sniff(iface='WI-FI', prn=packet_callback,
    filter="ip", store=0)
```

Source IP : 192.168.1.5

Destination IP : 172.217.16.142

---

protocol : ICMP

Source IP : 192.168.1.2

Destination IP : 8.8.8.8

---

protocol : UDP

Source IP : 192.168.1.10

Destination IP : 224.0.0.251

---

protocol : TCP

Source IP : 192.168.1.5

Destination IP : 172.217.16.142

Result:

Successfully executed code using RAW socket to implement packet sniffing